TSM 099

Homework 3

Common factors and advanced factoring

Topics 13-17 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

GCF first

1. Find the GCD of 72 and 108. Then find the greatest common monomial factor of 18x cubed and 30x squared.

2. Factor out the GCF.

$18x^3 - 30x^2$

Factor trinomials

3. Factor completely over the integers. Verify by multiplication.

$x^2 - 2x - 24$

4. Factor completely over the integers. Verify by multiplication.

$x^2 - 11x + 24$

5. Factor completely over the integers. Verify by multiplication.

$6x^2 + 7x + 2$

6. Factor completely over the integers. Verify by multiplication.

$8x^2 - 2x - 3$

7. Factor completely over the integers. Verify by multiplication.

$10x^2 - 13x + 3$

8. Factor completely over the integers. Verify by multiplication.

$12x^2 + 11x + 2$

Grouping and complete factorization

9. Factor by grouping.

$x^3 + 2x^2 - 9x - 18$

10. Factor by grouping.

$6x^3 - 9x^2 + 4x - 6$

11. Factor completely.

$3x^3 - 12x$

12. Factor completely.

$2x^2 - 8x + 8$

13. Factor completely.

$9x^2 - 25$

14. A student factors 4x squared - 16 as 4(x squared - 4) and stops. Explain why the answer is incomplete and finish it.